

Lab Safety

Mr. Queenan – Biology 9

Unit 1



Lab Safety

The “Why’s”

- <https://youtu.be/6aK2CKrdjbE>
- Why is lab safety the first lesson we are receiving as freshman in biology?
- Isn't lab safety obvious?
- What is the goal of this lesson?
 - To understand basic lab procedures
 - Relatively universal across all labs
- <https://www.youtube.com/watch?v=tsAHtoFiwNM>

The Golden Rule

Follow all instructions given by the lab manager and/or supervisors.



Basic Lab Safety

- **Be prepared** for an experiment or preparation by reading protocols before coming to the lab, when available
- Food and beverages are **not** allowed in the lab and are not to be stored in the lab refrigerator.
- Wear protective laboratory equipment, PPE (gloves, lab coat, goggles, etc.) when necessary.
- In the event of any emergency such as injury, spill, accident, fire or equipment failure, notify a lab supervisor **immediately**.
- Read all labels and safety instructions carefully before performing any lab tasks.
- Any protocol requiring hazardous chemicals is to be performed under the chemical fume hood, by trained laboratory personnel.

Basic Lab Safety

- Keep long hair tied back.
- Wear long pants, closed-toed shoes, and all other forms of clothing that will cover bare skin. Clothing should also not be loose or baggy, in order to avoid accidents.
- Never leave a hot plate that is in use unattended; assume the hot plate is hot and never touch the heating surface.
- Do not handle hot glassware with bare hands.
- Students are not permitted to work alone in the laboratory.

Basic Lab Safety

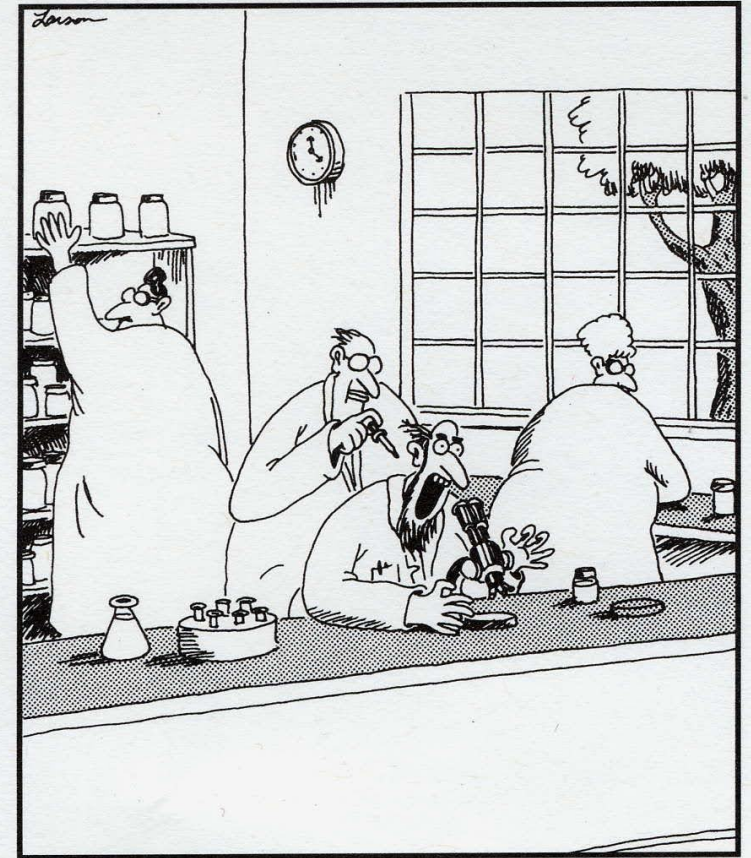
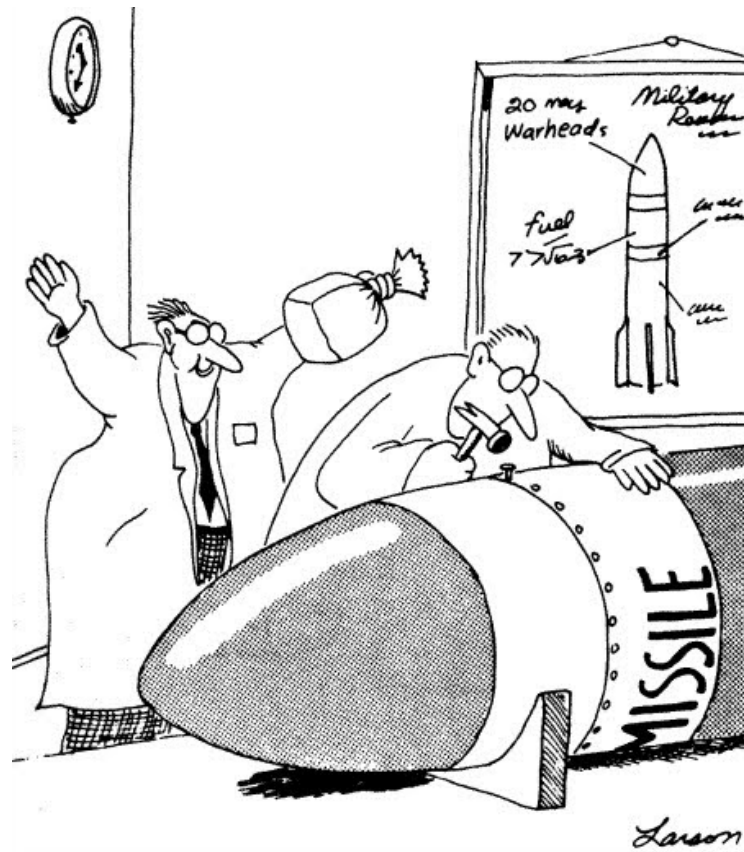


"My God! It is Professor Dickle! ...Weinberg, see if you can make out what the devil he was working on, and the rest of you get back to your stations."

Basic Lab Safety

- If any part of an experiment, preparation or equipment use is unclear, ask questions. Do not attempt to perform a task you do not understand.
- Do not leave an on-going experiment or preparation unattended.
- Keep aisles clear and store backpacks and jackets underneath tables to prevent tripping.
- No rough-housing or fooling around is allowed in the lab.
- Wash hands thoroughly when leaving the lab, especially before eating.

Basic Lab Safety



Professor Glickman, the lab practical joker, deftly places a single drop of hydrochloric acid on the back of Professor Bingham's neck.

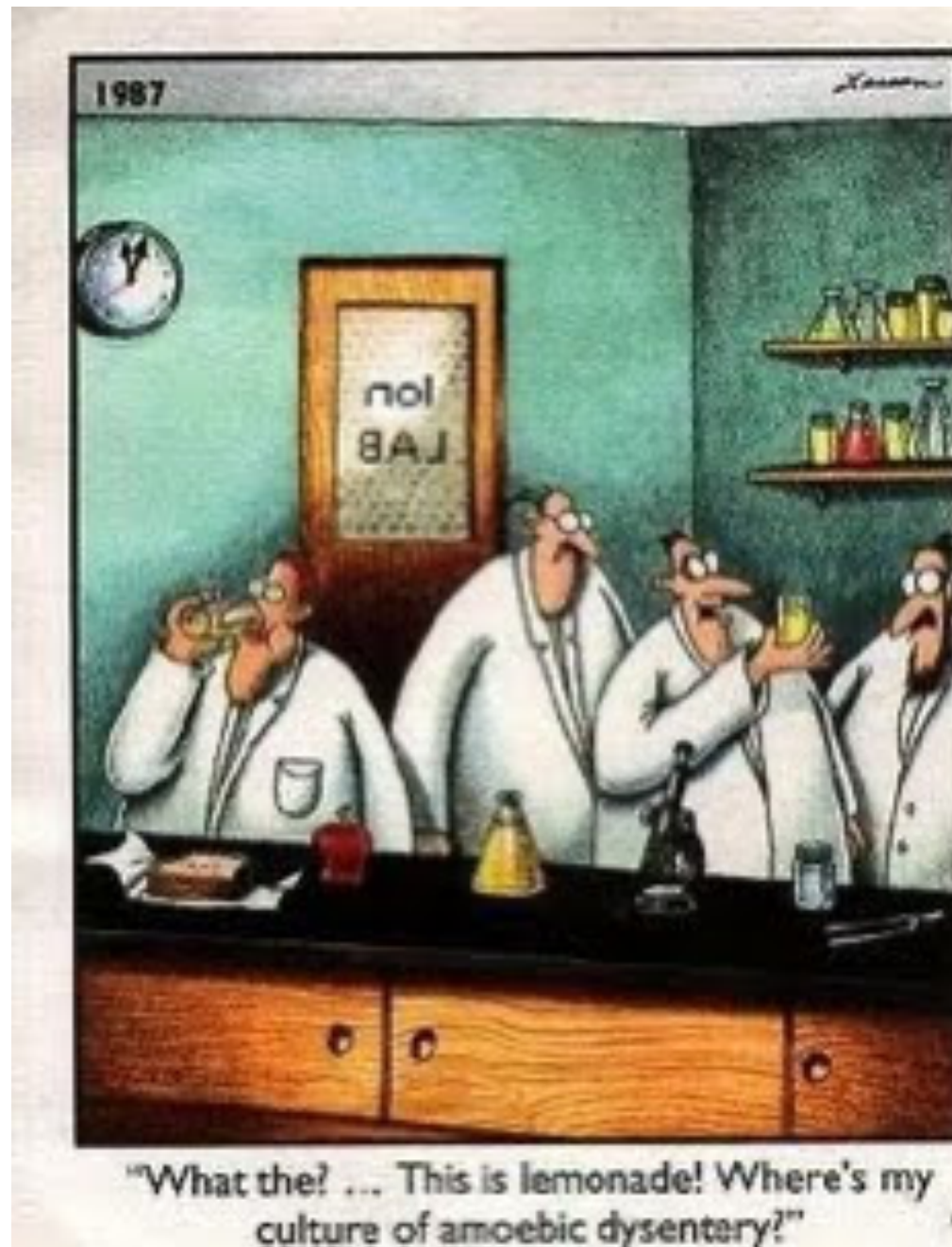
Emergency Procedures

- In case of:
 - Fire Drill
- Fire
- Broken Glass
- Power Failure
- Lockdown
- Accident (Bodily Harm)

Chemical & Sample Safety

- Never pipette any liquids by mouth; use a pipette pump.
- Any mixtures, solutions, or samples that are prepared by students or staff **MUST** be clearly labeled with:
 - All contents and concentrations.
 - Any warnings or precautions for use or storage.
 - Initials of the preparer.
 - Date prepared.
 - Expiration date (if known).
- Read all labels and warnings carefully before using any chemical. For further information, refer to the MSDS
 - Be sure to ask a lab supervisor if you have any questions about how to handle or dispose of a chemical.

Chemical & Sample Safety



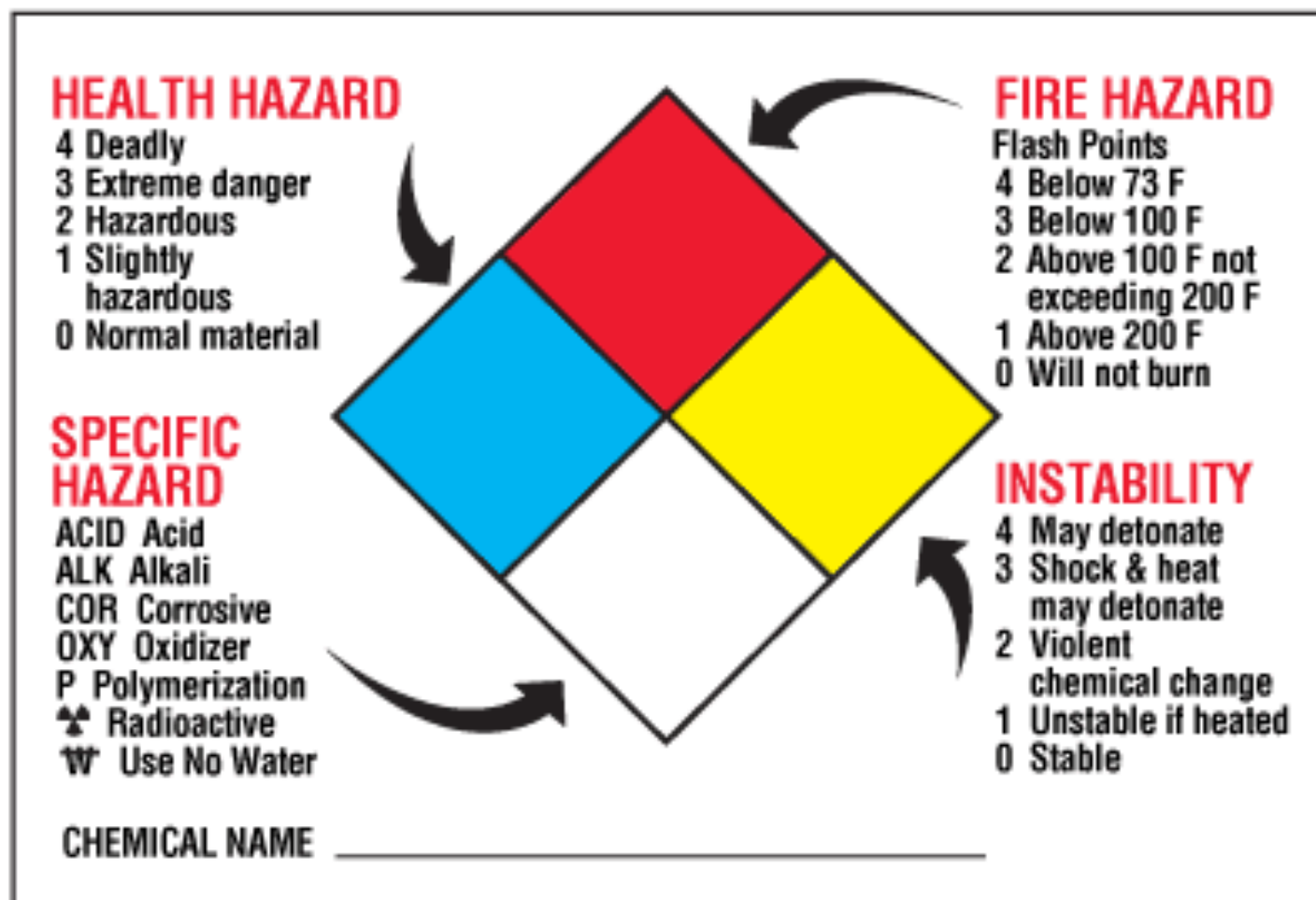
Chemical & Sample Safety

- Treat all chemicals as if they were dangerous; do not taste, touch or smell anything.
- Never assume that a clear liquid is water.
- Carry reagent bottles and beakers with two hands to avoid spilling or dropping them, or use a cart when available.
- Do not return unused reagent to the original container to avoid contamination.
- Do not leave containers and bottles open for extended periods of time. Close caps and lids tightly when not in use.
- Always return chemicals where you found them when you are finished.

Emergency Procedures

- In case of:
 - Chemical or sample splash to eyes
 - Spill
 - Skin or clothing contact with a hazardous substance
 - Ingestion
 - Inhalation

Safety Symbols



Safety Symbols

SAFETY PRACTICES



Flammable



Harmful /
Irritant



Corrosive



Poison /
Toxic



Explosion



Biohazard



Oxidizer



Environmental
Hazard



Radioactive



Laser Radiation